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Course: Calculus Beta

Assignment: MyLab Math - Uge 6

Find the Maclaurin series for the given function.

$$10 \cos(-x)$$

Let f be a function with derivative of all orders throughout some interval containing 0 as an interior point. Then the Maclaurin series of f is the Taylor series generated by f at $x = 0$.

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

First find $f(0)$.

$$f(x) = 10 \cos(-x)$$

$$\begin{aligned} f(0) &= 10 \cos(-0) \\ &= 10 \end{aligned}$$

Find $f'(x)$. Differentiate $f(x)$ with respect to x .

$$f(x) = 10 \cos(-x)$$

$$f'(x) = 10 \sin(-x)$$

Note that $10 \sin(-x) = -10 \sin x$ since the sine function is an odd function. Evaluate $f'(0)$.

$$f'(x) = -10 \sin x$$

$$\begin{aligned} f'(0) &= -10 \sin(0) \\ &= 0 \end{aligned}$$

Next find $f''(x)$. Differentiate $f'(x)$ with respect to x .

$$f'(x) = -10 \sin x$$

$$f''(x) = -10 \cos x$$

Evaluate $f''(0)$.

$$f''(x) = -10 \cos x$$

$$\begin{aligned} f''(0) &= -10 \cos(0) \\ &= -10 \end{aligned}$$

Find $f'''(x)$. Differentiate $f''(x)$ with respect to x .

$$f''(x) = -10 \cos x$$

$$f'''(x) = 10 \sin x$$

Evaluate $f'''(0)$.

$$f'''(x) = 10 \sin x$$

$$\begin{aligned} f'''(0) &= 10 \sin(0) \\ &= 0 \end{aligned}$$

So $f(x) = 10 \cos(-x)$, $f'(x) = -10 \sin x$, $f''(x) = -10 \cos x$, and $f'''(x) = 10 \sin x$. Notice that the even derivatives suggest a pattern and the odd derivatives suggest a pattern. Use these results to find a general formula for $f^{(2n)}(x)$ and for $f^{(2n+1)}(x)$. Start with $f^{(2n)}(x)$.

$$f^{(2n)}(x) = 10(-1)^n \cos x$$

Now find a general formula for $f^{(2n+1)}(x)$.

$$f^{(2n+1)}(x) = 10(-1)^{n+1} \sin x$$

Next, find a general formula for $f^{(2n)}(0)$ and $f^{(2n+1)}(0)$. This can either be done by identifying a pattern in the values of $f(0)$, $f'(0)$, $f''(0)$, and $f'''(0)$ or by substituting 0 for x in the general formulas for the derivatives. Here, we will substitute 0 for x in the general formula for the derivatives. Start with $f^{(2n)}(0)$.

$$f^{(2n)}(x) = 10(-1)^n \cos x$$

$$\begin{aligned} f^{(2n)}(0) &= 10(-1)^n \cos(0) \\ &= 10(-1)^n \end{aligned}$$

Next find $f^{(2n+1)}(0)$.

$$f^{(2n+1)}(x) = 10(-1)^{n+1} \sin x$$

$$\begin{aligned} f^{(2n+1)}(0) &= 10(-1)^{n+1} \sin(0) \\ &= 0 \end{aligned}$$

So at $x=0$, $\cos x$ is 1 and $\sin x$ is 0. This means that only the even powers of x will occur in the Maclaurin series for this function.

Use the fact that $f^{(2n)}(0) = 10(-1)^n$ and that only the even powers of x will occur to write the Maclaurin series for this function.

$$\sum_{k=0}^{\infty} \frac{10(-1)^k (x)^{2k}}{(2k)!}$$

Therefore, $\sum_{k=0}^{\infty} \frac{10(-1)^k (x)^{2k}}{(2k)!}$ is the Maclaurin series for $f(x) = 10 \cos(-x)$.